

Chandrasekhar's white dwarf equation, ADM with 11 terms

$\theta(\xi)$

Numerical Method / Constant (C)

RK5(4) / 0

ADM / 0

RK5(4) / 0.2

ADM / 0.2

RK5(4) / 0.4

ADM / 0.4

RK5(4) / 0.6

ADM / 0.6

RK5(4) / 0.8

ADM / 0.8

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) + (\theta^2 - C)^{\frac{3}{2}} = 0, \theta(0) = 1, \theta'(0) = 0$$

